

5-state EKF: a' as a measured state (y_2 update, no bolt-on)

DRAFT — route follows `5state_aprime_var_esti.tex`; Sec. 2 replaces the β -EWMA of a'_{hdm} with a Kalman update of a' .

0. Definitions

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k], \quad \Delta H_d[k] = h_d[k] - h_d[k-d]$$

$$a_d[k] := a_h(h_d[k]), \quad a'[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_r[k] := a_h(h[k]) - a_d[k]$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{ad}[k] - \varepsilon_h[k]$$

a_d = gain at the desired height (deterministic, trajectory-driven); a' = its slope; a_r = residual gain from the tracking error; ΔH_d = known d -step commanded sweep. $\delta h_1 := \delta h[k-d]$ (oldest, measured), $\delta h_2 := \delta h[k-1]$, $\delta h_3 := \delta h[k]$.

1. True

First-order Taylor of the gain at the actual (perturbed) height:

$$a_h(h[k]) = a_h(h_d[k] - \delta h[k]) \approx a_d[k] - a'[k] \delta h[k]$$

$$a_r[k] = -a'[k] \delta h[k], \quad a'[k+1] \approx a'[k]$$

The desired-height gain drifts with the commanded sweep:

$$a_d[k+1] = a_d[k] + a'[k] \Delta h_d[k]$$

The second measurement reads the *actual* gain d steps ago. Substituting $a_d[k-d] = a_d[k] - a' \Delta H_d[k]$ (from the sweep, a' slow) and $\delta h[k-d] = \delta h_1$:

$$y_2[k] = a_h(h[k-d]) = a_d[k-d] - a' \delta h[k-d] + \nu_2[k]$$

$$y_2[k] = a_d[k] - a'[k] (\Delta H_d[k] + \delta h_1[k]) + \nu_2[k]$$

a' enters y_2 *explicitly*, multiplying the total height offset $\Delta H_d + \delta h_1$ (commanded sweep + thermal tracking error). This is the relation the var-channel used implicitly; here it is kept as a measurement.

2. Estimator (a' inside the KF)

State (5×1):

$$x = [\delta h_1, \delta h_2, \delta h_3, a_d, a']^\top$$

Predict (deterministic map; $\Delta h_d, \Delta H_d$ known from the trajectory):

$$\begin{aligned} \delta h_1[k+1] &= \delta h_2[k], & \delta h_2[k+1] &= \delta h_3[k], & \delta h_3[k+1] &= \lambda_c \delta h_3[k], \\ a_d[k+1] &= a_d[k] + \hat{a}'[k] \Delta h_d[k], & a'[k+1] &= a'[k]. \end{aligned}$$

Measurement (y_1 = delayed position error, y_2 = gain gauge):

$$y_1[k] = \delta h_1[k] + n_x[k], \quad y_2[k] = a_d[k] - a'[k](\Delta H_d[k] + \delta h_1[k]) + \nu_2[k]$$

$$H_1 = [1, 0, 0, 0, 0]$$

$$H_2[k] = [-\hat{a}'[k], 0, 0, 1, -(\Delta H_d[k] + \delta \hat{h}_1[k])]$$

Update (standard EKF; $H = [H_1; H_2]$, $R = \text{diag}(\sigma_n^2, R_2)$):

$$K[k] = P^- H^\top (H P^- H^\top + R)^{-1}, \quad x^+ = x^- + K(y - Hx^-)$$

Why a' is now genuinely in the KF. The gain-row of K is

$$K_5 = \frac{[P^- H^\top]_5}{S} = \frac{P_{55}^- H_2(5) + P_{54}^- H_2(4) + \dots}{S}, \quad H_2(5) = -(\Delta H_d + \delta \hat{h}_1) \neq 0.$$

a' carries its *own* measurement entry $P_{55}^- H_2(5)$ (direct correction), not only the cross-covariance P_{54}^- . This replaces the esti recursion $a'[k+1] = \beta a'[k] + (1-\beta) a'_{hdm}[k]$ (a β -EWMA of a variance ratio computed *outside* the filter): a' now updates from the y_2 innovation like any other state.

3. Error dynamics

$e_h = \delta h - \delta \hat{h}$, $e_{a_d} = a_d - \hat{a}_d$, $e_{a'} = a' - \hat{a}'$. Predict (before correction):

$$e_{a_d}[k+1] = e_{a_d}[k] + \Delta h_d[k] e_{a'}[k], \quad e_{a'}[k+1] = e_{a'}[k],$$

so an a' error feeds the level error at rate Δh_d (the sweep coupling). Linearizing the y_2 innovation about the current estimates:

$$\nu_2 = e_{a_d} - (\Delta H_d + \delta \hat{h}_1) e_{a'} - \hat{a}' e_{h,1} + \nu_2^{\text{noise}}.$$

The closed loop is the usual EKF contraction

$$e[k+1] = (I - K[k]H[k]) F_e[k] e[k] + q[k]$$

with $e = [\delta h_1, \delta h_2, \delta h_3, e_{a_d}, e_{a'}]^\top$. Because $H_2(5) \neq 0$, the correction $(I - KH)$ acts on $e_{a'}$ *directly* through the y_2 innovation — the mechanism the bolt-on lacked.

4. F_e and H

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & 0 & 1 & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\hat{a}'[k] & 0 & 0 & 1 & -(\Delta H_d[k] + \delta \hat{h}_1[k]) \end{bmatrix}$$

Time-varying entries: $F_e(3,4) = -F_{dh}[k]$, $F_e(4,5) = \Delta h_d[k]$ (sweep coupling, predict), and the whole of $H_2[k]$ (measurement coupling). Row 5 of F_e is a pure integrator; a' acquires dynamics only through the measurement, not a process pole.

5. Observability and caveats

Two channels in one entry. $H_2(5) = -(\Delta H_d + \delta \hat{h}_1)$: the commanded sweep ΔH_d (deterministic, present only in motion) and the thermal tracking error $\delta \hat{h}_1$ (stochastic, present in holds). Both routes to a' live in the single measurement entry.

Bilinear EKF. y_2 contains the product $a' \delta h_1$; both $H_2(1) = -\hat{a}'$ and $H_2(5) = -(\Delta H_d + \delta \hat{h}_1)$ come from it, so a' and δh_1 are coupled in the same measurement. Identifiability (does an innovation load onto $e_{a'}$ or $e_{h,1}$?) is set by the covariance and must be checked — in a pure hold ($\Delta H_d = 0$) both are read from the same thermal product.

Hold floor unchanged. $\Delta H_d = 0 \Rightarrow H_2(5) = -\delta \hat{h}_1$, so a' observability scales with the thermal amplitude $\sim \sqrt{\text{Var}(\delta h)}$ over a finite dwell — a CRLB floor, independent of this reformulation.

Not self-referential. $y_2 = a_{xm}$ is a real measurement carrying $-a'_{\text{true}} \delta h$; the innovation delivers genuine truth to a' , unlike the var-channel's self-constructed $\hat{a}_r$ (which returns $\hat{a}'$).

6. Process / measurement noise (Q, R) and the closed loop

Predict noise $Q = \text{diag}(0, 0, Q_{33}, Q_{44}, Q_{55})$ (rows 1, 2 are pure delays), measurement noise $R = \text{diag}(R_1, R_2)$ on (y_1, y_2) . Four of the five nonzero entries are fixed by first principles; only Q_{55} carries a design choice (Option A below).

Q_{33} — **thermal drive of δh_3 .** The white term in $\delta h_3[k+1] = \lambda_c \delta h_3[k] - F_{dh} e_{a_d} - \varepsilon_h[k]$ is the thermal position increment $a f_T$ over one step. With $\text{Var}(f_T) = 4k_B T \gamma_N / \Delta t$ and $a = \Delta t / \gamma_N$,

$$\text{Var}(\varepsilon_h) = a^2 \text{Var}(f_T) = 4k_B T \frac{\Delta t}{\gamma_N} = 4k_B T a.$$

$$Q_{33} = 4k_B T \hat{a}_d$$

(fluctuation–dissipation; evaluated at the estimated gain $\hat{a}_d$).

R_1 — y_1 **sensor noise.** $y_1 = \delta h_1 + n_x \Rightarrow R_1 = \sigma_n^2$.

R_2 — $y_2 = a_{xm}$ **gauge noise.** a_{xm} is the IIR variance-ratio gain gauge (paper 2025, Eq. 9–13); its linear-inversion readout noise was derived as a quadratic in the gain,

$$R_2 = a_{cov} \text{IF}_{var} (\hat{a}_d + \xi)^2, \quad \xi = \frac{C_n}{C_{dpmr}} \frac{\sigma_n^2}{4k_B T},$$

with $C_{dpmr} = 2 + \frac{1}{1-\lambda_c^2} \approx 3.96$, $C_n = \frac{2}{1+\lambda_c} \approx 1.18$, $\text{IF}_{var} \approx 4.24$ ($\lambda_c = 0.7$, Option A autocorrelation). No delay-transport (+5 Q_{77}) term is needed: the d -step delay is carried explicitly by ΔH_d inside H_2 , not by an inflated R_2 .

Q_{44} — a_d (**structural**, ≈ 0). The predict $a_d[k+1] = a_d[k] + \hat{a}' \Delta h_d$ is first-order exact. Its two residuals are (i) the gain-estimate error $(a' - \hat{a}') \Delta h_d$, routed through $e_{a'}$ by $F_e(4, 5) = \Delta h_d$ (not process noise), and (ii) the Taylor truncation $\frac{1}{2} a'' \Delta h_d^2$, the only genuine model noise:

$$Q_{44} = \left(\frac{1}{2} a'' \Delta h_d^2 \right)^2 \approx 0 \quad (\Delta h_d = O(v_d \Delta t) \text{ per step}).$$

There is deliberately *no* $(\hat{a}')^2 Q_{33}$ term: the thermal gain wander lives in the reconstruction $\hat{a}_z = a_d - \hat{a}' \delta \hat{h}_3$ (a deterministic function of the state δh_3), not in the a_d process noise — this is the $\sim 30\times$ predict-side reduction over the leak-free-integrator gain state.

Caveat (code vs derivation, 2026-07-22). The temp harness currently runs $Q_{44} = (\hat{a}')^2 Q_{33}$ (the taylor gain-wander, a leftover from the slot-4=actual-gain parametrization), *not* this ≈ 0 form — i.e. the thermal wander is double-counted (once here, once in the reconstruction). Whether $Q_{44} \approx 0$ (this derivation) or $(\hat{a}')^2 Q_{33}$ (code) is correct is *unresolved*: the a' -uncertainty is already propagated into P_{44} by $F_e(4, 5) = \Delta h_d$, which argues for ≈ 0 . All §8/§9 numbers were run on the code's $(\hat{a}')^2 Q_{33}$.

Q_{55} — a' (**Option A**). a' is the slope *at the desired height* h_d , so it drifts only with the commanded sweep — no thermal term:

$$w_{a'}[k] = a'[k+1] - a'[k] = a''(h_d) \Delta h_d[k] \implies Q_{55}[k] = (a'' \Delta h_d[k])^2.$$

Hold ($\Delta h_d = 0$): $Q_{55} = 0$ *exactly* — a' is genuinely constant, P_{55} does not inflate from process noise, and the cap/gate that a flat Q_{55} would demand is unnecessary. The low-excitation hold is handled by the model, not an external switch.

$$\boxed{Q_{55}|_{\text{hold}} = 0, \quad Q_{55}|_{\text{motion}} = (a'' \Delta h_d)^2.}$$

Motion: $(a'' \Delta h_d)^2$ needs a'' , unknown (second derivative of the unknown $c(\bar{h})$). **Option A** takes $a'' \approx 0 \Rightarrow Q_{55} \approx Q_{55}^{\min}$ (a small floor keeping the channel alive); a' is then a constant state driven only by the y_2 (and forthcoming $y_3 = \delta x_{md}$) innovations. Zero assumption on c ; if a fast descent shows a' lag, escalate to adaptive $\hat{a}'' \approx \Delta \hat{a}' / \Delta h_d$ (Option B) or an a_d -curvature bound (Option C).

7. Identifiability and the closed loop

y_2 carries the product $a' \delta h_1$; $H_2(1) = -\hat{a}'$ and $H_2(5) = -(\Delta H_d + \delta \hat{h}_1)$ both come from it, so a' and δh_1 share the y_2 innovation and the split is set by P^- . In a pure hold ($\Delta H_d = 0$) both are read from the same thermal product, so a' observability scales as $\sqrt{\text{Var}(\delta h)}$ over the dwell — a CRLB floor. This floor *is* the model's own gentle behaviour: $K_5 \propto P_{55} H_2(5) \rightarrow 0$ as $\Delta H_d \rightarrow 0$ and $Q_{55} \rightarrow 0$, i.e. the “no gate” mechanism written in equations.

The closed loop is

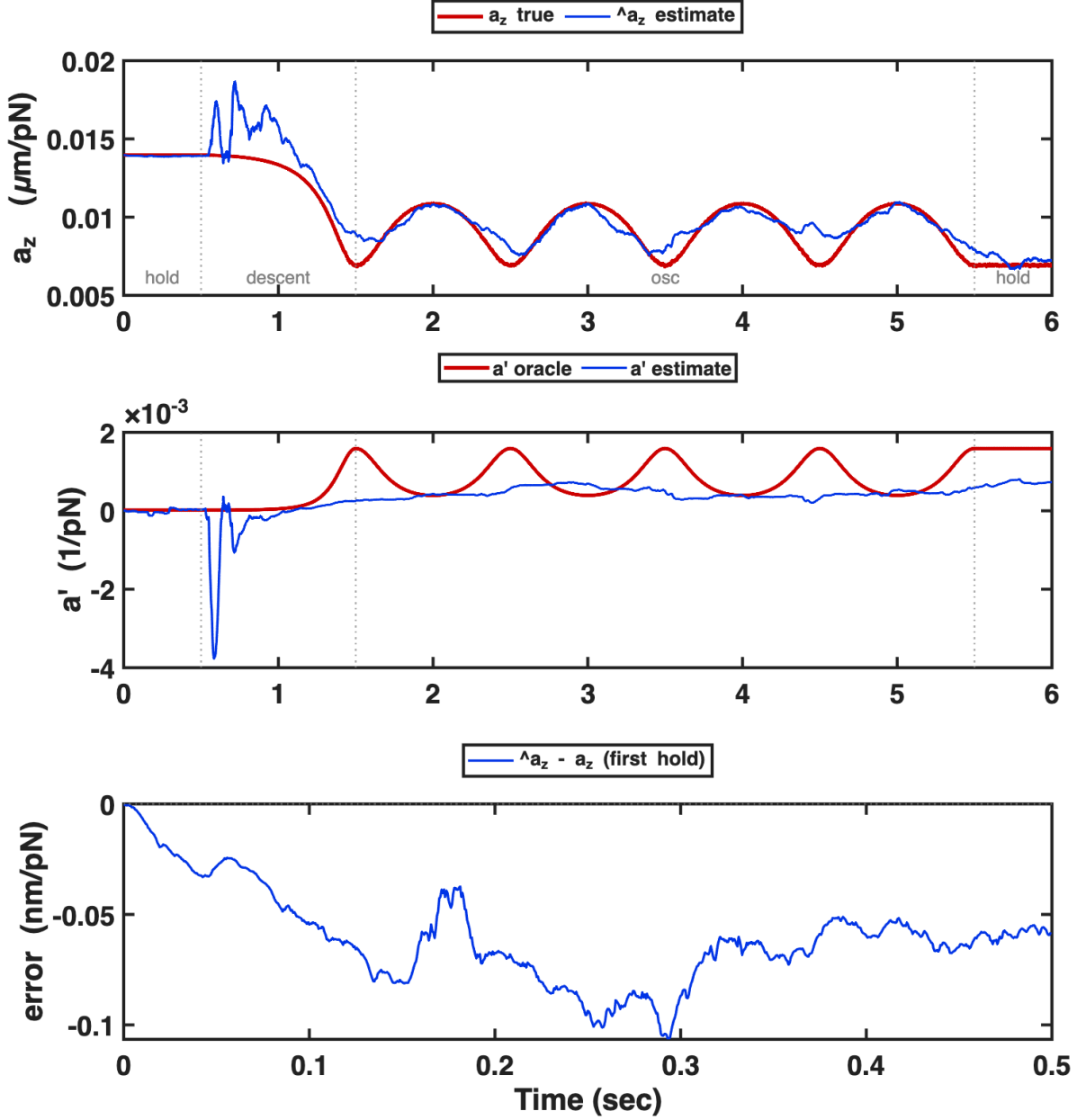
$$e[k+1] = (I - K[k]H[k])F_e[k]e[k] + q[k], \quad e = [\delta h_1, \delta h_2, \delta h_3, e_{a_d}, e_{a'}]^T,$$

with $Q = \text{diag}(0, 0, Q_{33}, Q_{44}, Q_{55})$ and $R = \text{diag}(R_1, R_2)$. Under Option A ($Q_{44}, Q_{55} \rightarrow \text{floor}$) the steady state solves the discrete Lyapunov/Riccati recursion; P_{55}^{ss} is the a' variance floor and, through $\hat{a}_z = a_d - \hat{a}' \delta \hat{h}_3$, sets the $\hat{a}_z$ floor. The full symbolic solve is deferred to the integration step (block D, with the deterministic channel $y_3 = \delta x_{md}$ added), since y_3 changes the a_d/a' observability.

8. Verification (Option A, 1 Hz canonical, pure-MATLAB)

Setup: $\bar{h}$ swept $22 \rightarrow 2$ over hold 0.5 s $\rightarrow$ descent 1 s $\rightarrow$ four 1 Hz cycles ($T = 6$ s), thermal + measurement noise on, seed 1 (+6-seed robustness). $Q_{55} = \text{Option A}$ ($a'' \approx 0$, floor 10^{-12}); the baseline reference is the κ -model $Q_{55} = \kappa^2 (\hat{a}')^2 \Delta h_d^2 / (h_d - h_w)^4$ (which implicitly assumes $a'' \sim \kappa a / (h_d - h_w)^2$, a mild c -form). Estimate $\hat{a}_z = a_d - \hat{a}' \delta \hat{h}_3$.

metric	Option A	baseline κ	oracle (a' known)
first-hold $\text{std}(\hat{a}_z)$ [nm/pN]	0.021	0.022	0.015
osc rel. err ($\hat{a}_z$)	6.5%	5.6%	2.3%
osc $\text{corr}(a', a'_{\text{oracle}})$	≈ 0	≈ 0	—
$\max a' $ (6 seeds)	$< 4 \times 10^{-3}$	$< 4 \times 10^{-3}$	—



Top: gain $\hat{a}_z$ (blue) vs a_z true (red). Middle: a' estimate vs oracle. Bottom: first-hold error $\hat{a}_z - a_z$ (< 0.1 nm/pN).

Confirmed. (i) First-hold $\hat{a}_z$ is clean ($0.021 \approx 0.015$ nm/pN, *no gate*): with $\Delta h_d = 0$ the a_d predict freezes and $Q_{55} = 0$, so P_{55} does not inflate and a' stays near its (correct) ≈ 0 — the “no gate” claim of §6–7 realized. (ii) No catastrophic blow-up across 6 seeds ($\max |a'| < 4 \times 10^{-3}$); the earlier reported $a' \rightarrow -24$ was a sporadic first-run EKF fragility, not reproduced. (iii) Option A (charter-clean, $a'' \approx 0$) is marginally smoother but $\sim 1\%$ less accurate than the κ -model (charter-questionable); Option A is kept.

Two residuals (open). (a) *Descent-onset transient* — at the descent start a' over-corrects once ($\rightarrow -3.8 \times 10^{-3}$), so $\hat{a}_z$ overshoots $\sim 30\%$ for ~ 0.4 s then recovers (bounded, transient phase). Cause: P_{55} is held at its prior through the info-starved hold, then dumped by the first strong descent innovation. Fix is a rate-limit on a' or a smaller P_{55}^0 (not a gate). (b) *Osc accuracy floor* — $\hat{a}_z$ osc error 6.5% vs oracle 2.3%, and a' itself is not tracked (osc corr ≈ 0): the single gain gauge a_{xm} is too noisy to pin a' under fast motion. Breaking this needs the independent deterministic channel $y_3 = \delta x_{md}$ (C2, block C), not a Q/R retune.

9. The deterministic channel C_2 ($y_3 = \delta x_{md}$, level measurement)

The discarded half. The tracking error splits as $\delta x_m = \delta x_{md} + \delta x_{mr}$ — a deterministic (systematic) part and a random (thermal) part. The gauge $y_2 = a_{xm}$ uses only $\text{Var}(\delta x_{mr})$, throwing δx_{md} away. C_2 recovers it as an *independent* measurement.

Mechanism. The controller applies $f_d = (1/\hat{a}_z)\{\dots\}$ with the reconstructed gain $\hat{a}_z = \hat{a}_d - \hat{a}' \delta \hat{h}_3$; a gain mismatch injects a deterministic forcing, and the systematic error obeys (Sec. 0)

$$\delta x_{md}[k+1] = \lambda_c \delta x_{md}[k] - F_{dh}[k] (a - \hat{a}_z) + O(e^2), \quad F_{dh}[k] = f_d[k] + (1 - \lambda_c) \sum_i f_d[k-i].$$

To first order the reconstruction error is the *level* error, $a - \hat{a}_z = (a_d - \hat{a}_d) - (a' \delta h - \hat{a}' \delta \hat{h}_3) = e_{a_d} + O(e^2)$ (the a' term is a product of errors), so C_2 measures the level state a_d . F_{dh} (the applied control force) is known and c -free — and is exactly the $F_e(3, 4)$ coupling already in the model.

Extraction $\rightarrow$ measurement (parallel to y_2): low-pass both sides and solve for the level error,

$$y_3[k] := \delta x_{md}[k+1] - \lambda_c \delta x_{md}[k] = -F_{dh}[k] e_{a_d}[k] + \nu_3[k], \quad z_3 := \hat{a}_d - \frac{y_3}{F_{dh}} = a_d.$$

z_3 is a direct, c -free measurement of the true desired-height gain:

$$H_3 = [0, 0, 0, 1, 0] \text{ (on } a_d), \quad \kappa_{C_2} := \text{recovery efficiency} \approx 0.90\text{--}0.94 \text{ (unbiased)}.$$

Where y_2 reads the *delayed* gain through a variance ($H_2 = [-\hat{a}', 0, 0, 1, -(\Delta H_d + \delta \hat{h}_1)]$), y_3 reads the *current level* through the deterministic residual — a physically independent route.

R_3 (route C: white).

$$R_3[k] = \frac{\sigma_{\nu_3}^2}{F_{dh}^2[k]}, \quad |F_{dh}[k]| < F_{\min} \Rightarrow R_3 \rightarrow \infty \text{ (gate off)}.$$

ν_3 is the random part δx_{mr} leaking through the finite-window low-pass. Verified floor: one W -window recovers a_d to $\sim 9.4\% a$ at $W = 0.5$ s ($\sim 1/\sqrt{W}$), and to $6.2\% a$ in the descent (single-direction excitation — the golden segment); a hold is uninformative ($\sim 26\%$ at low force), handled by the $F_{\min}$ gate. Route C books ν_3 white; it is in fact EWMA-coloured (same family as coloured- a_{xm}), deferred to an AR(1) honesty pass if P_{44} over-confidence shows.

Why it passes the a_{xm} floor. ν_3 (deterministic-residual extraction) is independent of the y_2 noise (variance-ratio), so two gain measurements combine as $1/\sigma^2 = 1/\sigma_{y_2}^2 + 1/\sigma_{y_3}^2$. They are time-complementary: the descent/oscillation, where a_{xm} is noisiest, is exactly where y_3 is sharpest, so the combined level variance drops below the a_{xm} -only floor. In a hold both channels gate off (no excitation) — consistent with the CRLB, no free lunch.

Caveats. (i) *Endogeneity*: F_{dh} must be the deterministic applied force F_{dh}^{det} (low-passed), not the instantaneous f_d , or y_3 correlates with its own noise and biases. (ii) *Slow*: window-integrated (0.5–1 s), ≥ 4 windows for 5% — a level trimmer, not a per-step corrector. (iii) *Honesty*: the white R_3 is provisional (route C); check P_{44} before trusting the gain-error bar.

Verification (route C, 1 Hz canonical, pure-MATLAB). Adding y_3 to the Option-A filter (per-step, white R_3) cuts the descent rel. error of $\hat{a}_z$ from 12.2% to 6.1% (halved, robust across seeds) — the previously worst segment. The oscillation improves only modestly (8.5% $\rightarrow$ 7.5%): there F_{dh} oscillates through zero and the residual weakens (prior C_2 : descent 100% vs osc 85–93% alive). A block *multi-window* weighted-LS was tested and *rejected* — monotonically worse as the window grows (both segments), because the per-step update with $R_3 \propto F_{dh}^{-2}$ *already is* recursive weighted least-squares: the Kalman recursion auto-down-weights the zero-crossings, and an explicit window only adds staleness (a_d drifts within it) and fewer updates. Per-step is C2’s optimal form; the osc floor is its genuine limit, not a tuning artefact.